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Lab 06: Jenkins Live

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Objective

Run a real controller + agent with Docker Compose, connect them, and execute the canonical three-stage Jenkinsfile.

Prerequisites

- Docker; concepts: Jenkins Setup, Jenkins Pipelines.

Architecture

Compose: jenkins (controller, port 8080, volume jenkins_home) + agent (inbound, label docker-agent) → Multibranch or Pipeline job → Build/Test/Deploy stages.

Setup

```
mkdir jenkins-lab && cd jenkins-lab # docker-compose.yml: jenkins/jenkins:its + inbound agent joined with the secret from the controller UI
docker compose up -d
```

Get the admin password: `docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword`.

Step 1 — Connect the agent

Execute

Manage Jenkins → Nodes → create node agent-1, labels docker-agent, launch via inbound; run the agent container with the secret + controller URL. Install suggested plugins.

Expected Result

Node page shows agent-1 online with 2–4 executors.

Why

Controller schedules, agents execute — this split is the whole architecture (see Jenkins Architecture).

Step 2 — Run the canonical pipeline

Execute

New Pipeline job, agent docker-agent, script:

```
pipeline { agent { label 'docker-agent' } stages { stage('Build') { steps { sh 'echo build' } }  
stage('Test') { steps { sh 'echo test' } } stage('Deploy'){ steps { echo 'deploy' } } } }
```

Expected Result

Stage View shows three green boxes on agent-1.

Why

The minimal pipeline that proves scheduling, execution, and visualization all work.

Verification

- [] Agent online, labeled, executors idle after run.
- [] Three green stages in Stage View.
- [] Controller ran zero builds (check: no executors on master).

Failure Scenarios

- Agent offline: wrong secret/URL or controller unreachable — check container logs, use `http://jenkins:8080` in-Compose.
- No executors on controller is correct — never enable them in prod.

Cleanup

`docker compose down` keeps `jenkins_home` volume; `down -v` wipes everything.

What This Proved

Controller/agent split hands-on. Continue with Lab 07.

Interview Notes

- `down` vs `down -v`: which one wipes `jenkins_home` and why that distinction is the whole lesson.

- Zero executors on the controller — what you check to prove no build ever ran there.

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